Name:		Index Number:		Class:	

DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H2 PHYSICS

9478/01

22 September 2026

1 hour

Paper 1 Multiple Choice

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **thirty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct
- and record your choice in **soft pencil** on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and index number on the multiple choice answer sheet in the
- spaces provided unless this has been done for you.
- Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You may use an approved calculator.

INFORMATION

- The total mark for this paper is 30.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.

Data

speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
permeability of free space	$\mu_0 = 4\pi \times 10^{-7}\,\text{H m}^{-1}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$
	$(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1})$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
unified atomic mass constant	$u = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31}\,\text{kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27}\,\text{kg}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Avogadro constant	$N_A = 6.02 \times 10^{23}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on / by a gas	$W = p\Delta V$
pressure	$p = \frac{F}{A}$
gravitational potential	$\phi = -\frac{GM}{r}$
temperature	$T/\text{K} = T/\,^\circ\text{C} + 273.15$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$
mean translational kinetic energy of an ideal gas particle	$E = \frac{3}{2}kT$

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displacement of particle in s.h.m.	$x = x_0 \sin \omega t$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t = \pm \omega \sqrt{(x_0^2 - x^2)}$
electric current	$I = nAvq$
resistors in series	$R = R_1 + R_2 + ...$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + ...$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + ...$
capacitors in parallel	$C = C_1 + C_2 + ...$
energy in a capacitor	$U = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$
charging a capacitor	$Q = Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$
discharging a capacitor	$Q = Q_0 e^{-\frac{t}{\tau}}$
RC time constant	$\tau = RC$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
alternating current / voltage	$x = x_0 \sin \omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_0 nI$
energy states for quantum particle in a box	$E_n = \frac{h^2}{8mL^2}n^2$
radioactive decay	$x = x_0 e^{-\lambda t}$
radioactive decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

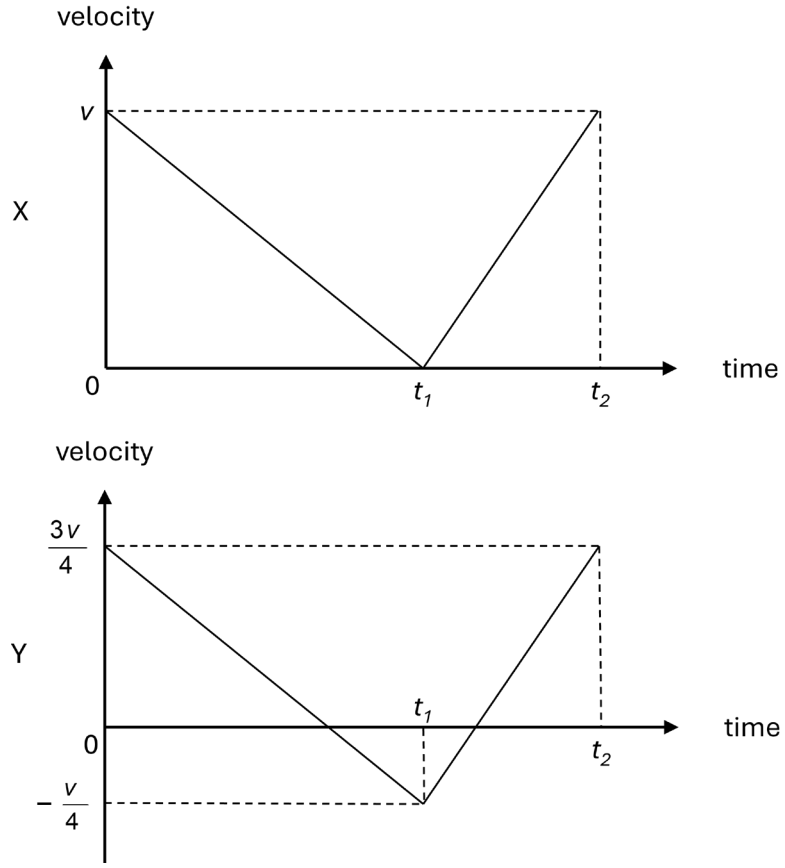
1 After taking measurements of the quantities in the expression $\frac{3PQ^4}{R^2}$, the total uncertainty is calculated as 11%.

Which individual percentage uncertainties of *P*, *Q* and *R* when combined give this total of 11%?

	***P* / %**	***Q* / %**	***R* / %**
A	1.0	1.0	2.0
B	2.0	2.0	1.5
C	3.0	1.0	2.0
D	5.0	2.0	1.0

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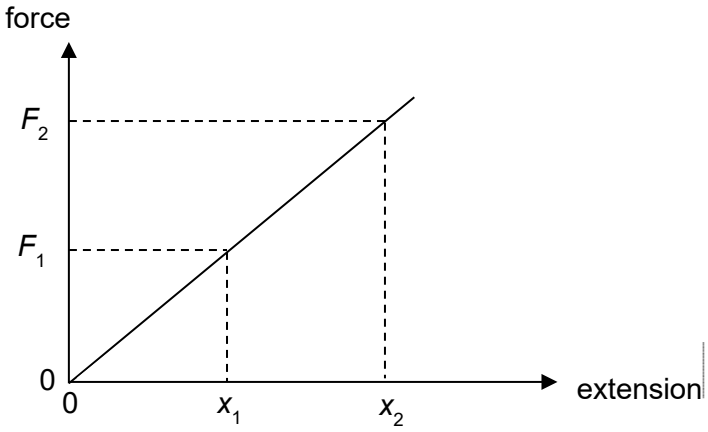
2 The motions of two objects X and Y along a straight line are shown in the following velocity-time graphs.



How do the magnitudes of the displacement and the total distance travelled compare for the two objects from time 0 to time t_2?

	displacement	**distance**
A	different	different
B	different	same
C	same	different
D	same	same

3 The force-extension graph of a spring is as shown.



What is the change in elastic potential energy when the extension of the spring is increased from x_1 to x_2?

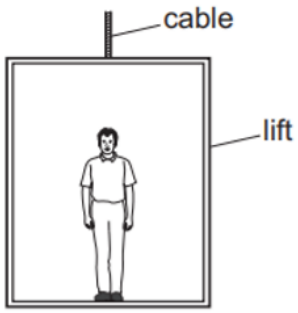
A $F_2x_2 - F_1x_1$

B $\frac{1}{2}(F_2x_2 - F_1x_1)$

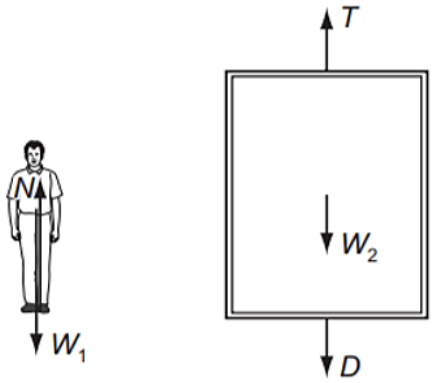
C $F_2x_2^{\,2} - F_1x_1^{\,2}$

D $\frac{1}{2}(F_2x_2^{\,2} - F_1x_1^{\,2})$

4 The diagram shows a man standing in a lift.



The forces acting on the man and the forces acting on the lift are shown below.



N is the force from the lift floor on the man.
W_1 is the weight of the man.
T is the tension in the lift cable.
W_2 is the weight of the lift.
D is the force from the man on the lift floor.

Which of the following statements is correct?

A $(W_1 + W_2)$ is always equal to T.

B If $N = W_1$, the lift must be at rest.

C N and W_1 are always equal and opposite.

D If $T = (D + W_2)$, the lift must have a constant velocity.

5 A trolley runs down a slope with a constant acceleration *a*. The mass of the trolley is now doubled, and the trolley is allowed to run down the same slope. In both cases effects of friction and air resistance are negligible.

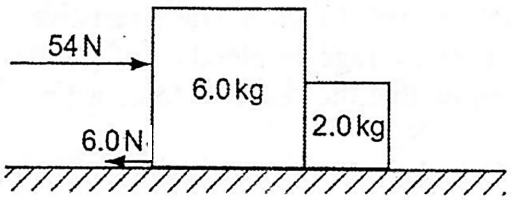
A The accelerating force is the same.

B The acceleration is $\frac{1}{2}a$.

C The acceleration is *a*.

D The acceleration is 2*a*.

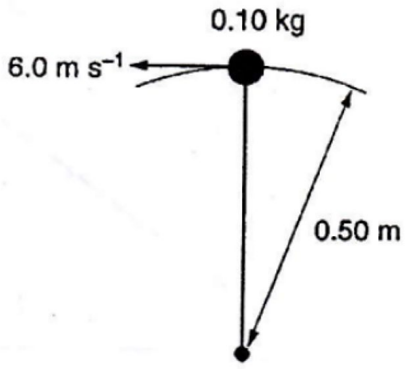
6 A force of 54 N pushes two touching blocks of mass 6.0 kg and 2.0 kg along a flat surface. The frictional force between the blocks and the surface is 6.0 N.



What is the force exerted on the 2.0 kg block?

A 12 N **B** 36 N **C** 45 N **D** 48 N

7 A ball of mass 0.10 kg is attached to a string and swung in a vertical circle of radius 0.50 m, as shown. Its speed at the top of the circle is 6.0 m s^{-1}.



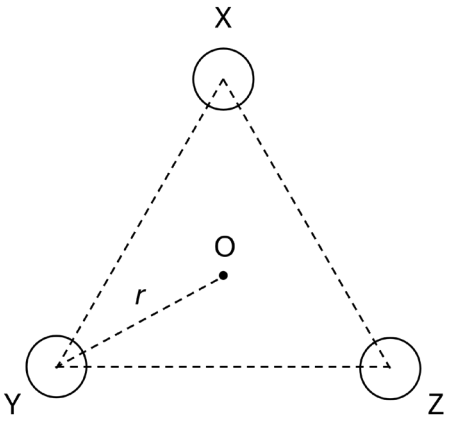
What is the tension in the string at this moment?

A 0.98 N **B** 6.2 N **C** 7.2 N **D** 8.2 N

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Use the following diagram to answer questions 8 and 9.

Three planets, X, Y and Z have masses *M*, *M* and *2M* respectively, are arranged in a triangle and are equidistant from each other.



8 What is the magnitude and direction of the gravitational field strength at point O?

	magnitude	**direction**
A	0	no direction
B	$\frac{GM}{r^2}$	towards X
C	$\frac{GM}{r^2}$	towards Z
D	$\frac{G\left(2M-\sqrt{3}M\right)}{r^2}$	towards Z

9 What is the gravitational potential at O and its direction?

	potential	**direction**
A	0	no direction
B	$-\frac{GM}{r}$	towards Z
C	$\frac{GM}{r}$	towards Z
D	$-\frac{4GM}{r}$	no direction

10 A small mass on a vibrating platform undergoes simple harmonic motion. The total energy of the mass is 18 mJ.

The amplitude of the oscillation is now doubled, and the time period is increased to be three times as long.

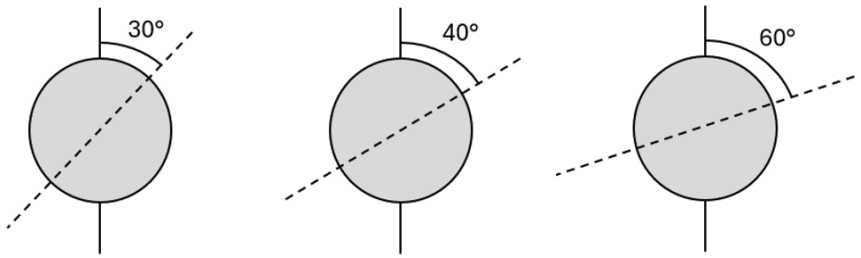
What is the new total energy of the mass?

A 8 mJ **B** 12 mJ **C** 36 mJ **D** 72 mJ

11 The diagram below shows three polarisers W, X and Y.

A beam of unpolarised light of intensity I_0 is incident normally onto the surface of polariser W, which then passes through polariser X and eventually emerges from polariser Y.

The axis of polarisation of each polariser is indicated by a dashed line.



What is the intensity of the light that emerges from polariser Y?

A $0.11\,I_0$ **B** $0.32\,I_0$ **C** $0.43\,I_0$ **D** $0.64\,I_0$

12 A light wave of amplitude *A* is incident normally on a surface of area *S*. The intensity reaching the surface is I.

The amplitude of the light wave is increased to 2*A*. The light is then focused on to a smaller area $\frac{1}{3}S$.

What is the intensity on this smaller area?

A $6I$ **B** $12I$ **C** $18I$ **D** $24I$

13 The diameter of the receiving dish of a telescope is 5.1 m. It receives light of wavelength $0.40\ \mu m$ from two point sources at a distance of 2.8×10^{25} m away.

What is the minimum separation of the two points sources so that the light from them can be resolved?

A 1.1×10^{18} m

B 2.2×10^{18} m

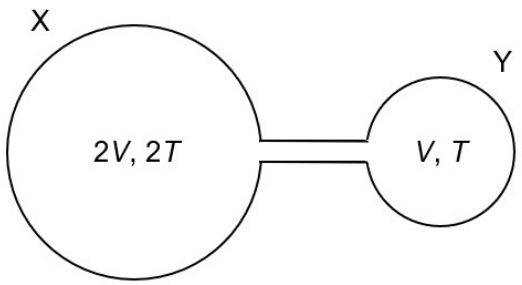
C 1.8×10^{32} m

D 3.6×10^{32} m

14 An ideal gas has thermodynamic temperature T, and its molecules has root-mean-square speed c_{rms}. If the gas is heated to temperature $2T$, what is the new root-mean-square speed of the molecules?

A $\sqrt{2}c_{rms}$ **B** $2c_{rms}$ **C** $2\sqrt{2}c_{rms}$ **D** $4c_{rms}$

15 An ideal gas is contained in two spherical containers X and Y of volume $2V$ and V respectively, connected by a hollow tube of negligible volume. The containers X and Y are maintained at temperatures $2T$ and T respectively. The setup is shown in the figure below.



Determine the ratio

$$\frac{\text{number of moles of gas in container X}}{\text{number of moles of gas in container Y}}.$$

A 0.25 **B** 1.00 **C** 2.00 **D** 4.00

16 A 100 g block of ice at −10.0 °C is placed into 150 g of water initially at 20.0 °C inside a perfectly insulated calorimeter. What is the final temperature of the mixture and the mass of ice remaining?

Specific heat capacity of water = 4180 J kg^{-1} K^{-1}

Specific latent heat of fusion of ice = 334000 J kg^{-1}

Specific heat capacity of ice = 2100 J kg^{-1} K^{-1}

A 0 K with 31.3 g of ice remaining

B 0 K with 68.7 g of ice remaining

C 0 °C with 31.3 g of ice remaining

D 0 °C with 68.7 g of ice remaining

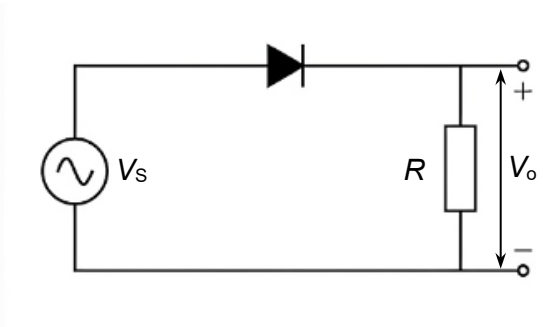
17 An electric current of 12 A is in a wire of length 0.35 m.

The average drift speed of the free electrons (charge carriers) in the wire is 5.0×10^{-4} m s^{-1}.

How many free electrons are in the wire?

A 1.3×10^{16} **B** 5.3×10^{22} **C** 1.5×10^{23} **D** 4.3×10^{23}

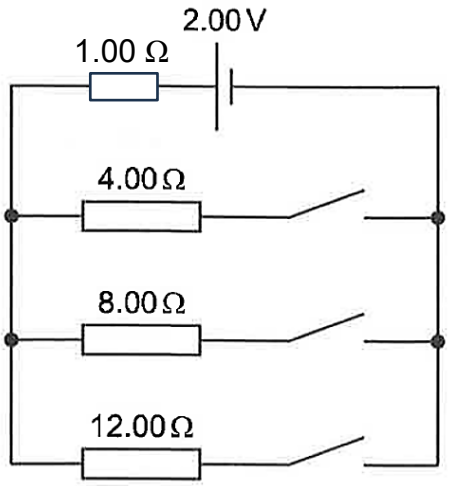
18 An ideal semiconductor diode is connected in series with a load resistor R = 50 Ω and an alternating voltage source $V_S = 12.0\sin(100\pi t)$. The output voltage V_o is measured across the load resistor R.



What is the average power dissipated in the load resistor R over one complete cycle?

A 0 W **B** 0.72 W **C** 1.44 W **D** 2.88 W

19 A cell of e.m.f. 2.00 V and internal resistance 1.00 Ω is connected to three resistors of 4.00 Ω, 8.00 Ω and 12.00 Ω as shown in the circuit diagram.



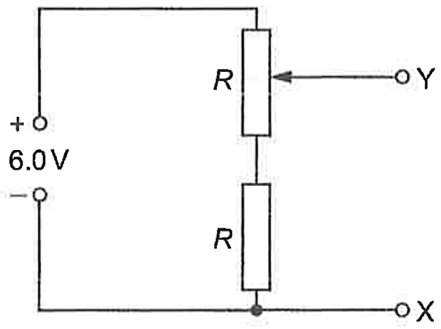
What is the current through the cell when every switch is closed?

A 0.0800 A **B** 0.500 A **C** 0.629 A **D** 0.917 A

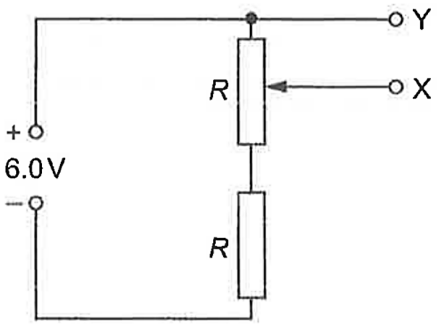
20 A potential divider has a constant supply of 6.0 V, as shown in the diagrams.

Which circuit will provide a potential difference between X and Y that can be varied between zero and 3.0V?

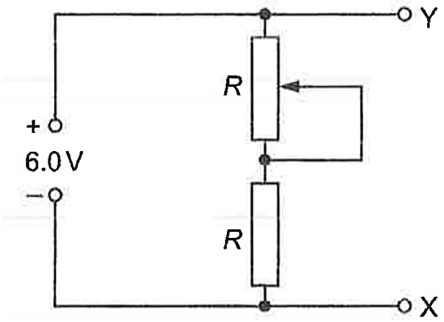
A



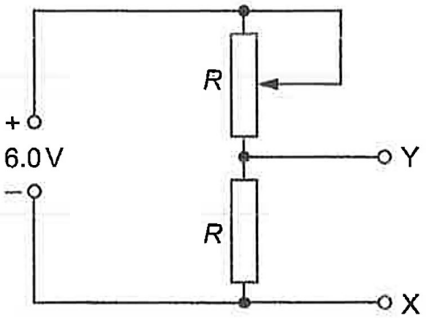
B



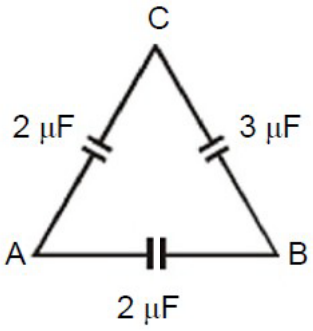
C



D



21 Three capacitors are connected in the shape of a triangle ABC as shown in the figure below.



A 5.0 V battery is connected between A and B.

What is the voltage between B and C?

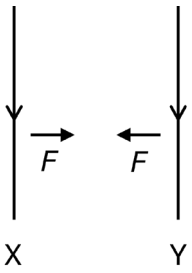
A 1.0 V **B** 2.0 V **C** 2.5 V **D** 3.0 V

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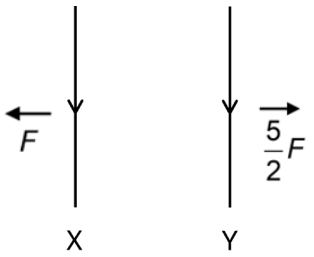
22 Two long, straight, current-carrying wires, X and Y, are parallel. The current in wire X is 2 A and the current in wire Y is 5 A.

Which diagram shows the correct combination of the magnitude of the forces and the directions of the current in wires X and Y?

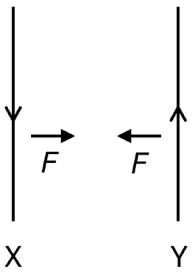
A



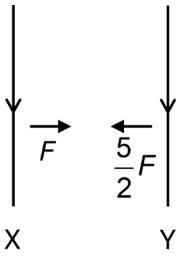
B



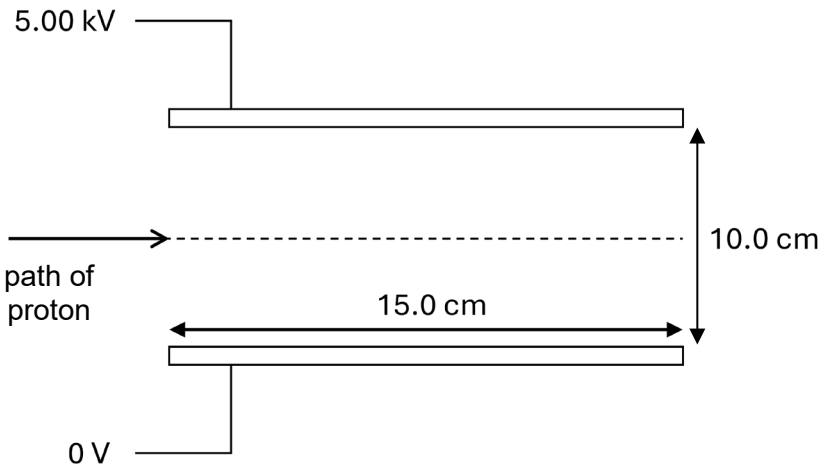
C



D



23 A proton, initially travelling horizontally at 2.52×10^6 m s^{-1}, enters a uniform electric field at right angles to the field. The electric field is generated by applying a potential difference of 5.00 kV between two parallel conducting plates 15.0 cm long and 10.0 cm apart.



What is the change in height of the proton as it leaves the plates and the direction which it is deflected?

`

	change in height / m	direction of deflection
A	8.5×10^{-3}	downwards
B	8.5×10^{-3}	upwards
C	15.6×10^{-3}	downwards
D	15.6×10^{-3}	upwards

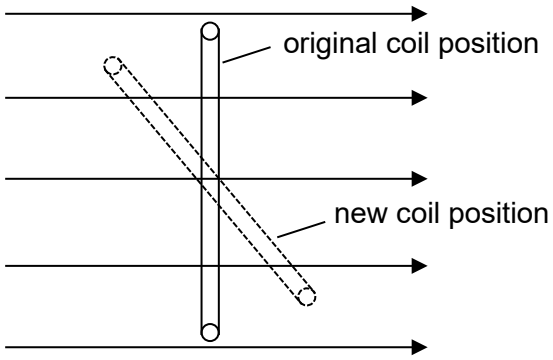
24 Two point charges each of charge Q at a separation of r repel each other with a force F.

Each charge is increased by adding $3Q$ and the separation is increased to $2r$.

What is the force between the charges?

A F B $9F/4$ C $4F$ D $8F$

25 A coil is situated with its plane perpendicular to a uniform magnetic field and within the field.



The coil is rotated to the new position shown.

How does the magnetic flux density and the total magnetic flux passing through the coil change during the rotation?

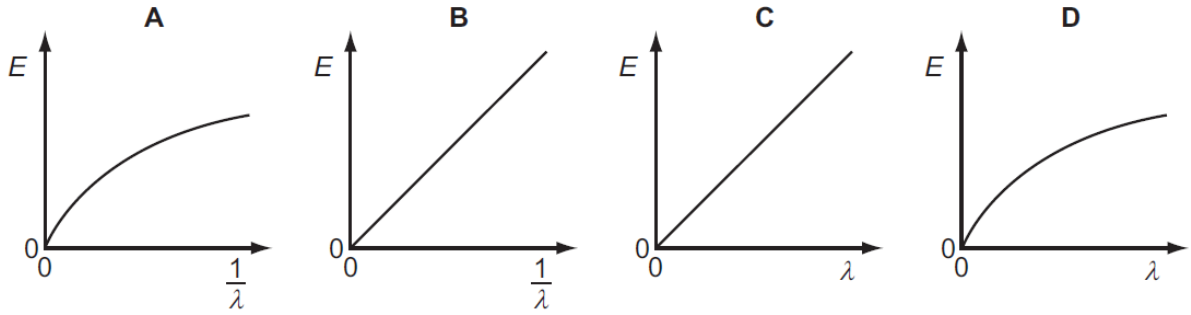
	magnetic flux density	**total magnetic flux**
A	decrease	decrease
B	increase	increase
C	no change	decrease
D	no change	increase

26 A coil produces a magnetic flux of 0.85 Wb. A second coil of 1500 turns is wound on the same core, such that all the magnetic flux from the first coil links the second coil.

What is the e.m.f. induced across the terminals of the second coil when the flux produced by the first coil is reduced uniformly to zero in 0.19 s?

A 0 V **B** 4.47 V **C** 335 V **D** 6710 V

27 Which graph shows the relationship between photon energy E and wavelength λ of electromagnetic waves?



28 A particle of mass m in a well of width L has ground-state energy E.

A second particle has mass $2m$ and is confined in a well of width $2L$.

Its ground-state energy is:

A $E/2$

B $E/4$

C $E/8$

D $E/16$

29 A sample of a radioactive isotope with nucleon number 210 decays by alpha emission into a stable isotope with nucleon number 206. The process has a half-life of 140 days.
If all the emitted alpha-particles escape from the sample but none of the daughter product escapes, what will be the percentage decrease in the mass of the source after 280 days?

A 0.48% **B** 1.43% **C** 25% **D** 75%

30 In medical radiotherapy, different types of radioactive emissions are chosen based on their penetrating abilities and ionising effects. Which combination correctly matches the emission type with its most appropriate medical application?

A alpha particles - external beam therapy for deep tumours

B beta particles - treating shallow skin cancers and surface tumours

C gamma rays - treating only surface-level conditions

D alpha particles - medical imaging of internal organs

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